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# The effects of remote work on collaboration among information workers

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## Abstract

The coronavirus disease 2019 (COVID-19) pandemic caused a rapid shift to full-time remote work for many information workers. Viewing this shift as a natural experiment in which some workers were already working remotely before the pandemic enables us to separate the effects of firm-wide remote work from other pandemic-related confounding factors. Here, we use rich data on the emails, calendars, instant messages, video/audio calls and workweek hours of 61,182 US Microsoft employees over the first six months of

2020 to estimate the causal effects of firm-wide remote work on collaboration and communication. Our results show that firm-wide remote work caused the collaboration network of workers to become more static and siloed, with fewer bridges between disparate parts. Furthermore, there was a decrease in synchronous communication and an increase in asynchronous communication. Together, these effects may make it harder for employees to acquire and share new information across the network.

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## Data availability

An anonymized version of the data supporting this study is retained indefinitely for scientific and academic purposes. The data are not publicly available due to employee privacy and other legal restrictions. The data are available from the authors on reasonable request and with permission from Microsoft Corporation.

## Code availability

The code supporting this study is retained indefinitely for scientific and academic purposes. The code is not publicly available due to employee privacy and other legal

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Contributions

L.Y. analysed the data. L.Y., D.H., S.J. and S. Suri performed the research design, interpretation and writing. S. Sinha, J.W., C.J., N.S. and K.S. provided data access and expertise. B.H. and J.T. advised and sponsored the project.

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## Ethics declarations

Competing interests

L.Y., S.J., S. Suri, S. Sinha, J.W., C.J., N.S., K.S., B.H. and J.T. are employees of and have a financial interest in Microsoft. D.H. was previously a Microsoft intern. All of the authors are listed as inventors on a pending patent application by Microsoft Corporation (16/942,375) related to this work.

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